Huawei Access Node YEL-ACC-52 QoS/Congestion Event SOP for Major Alarm ALARM_8979

Node: YEL-ACC-52 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_8979

Description

The Huawei Access Node YEL-ACC-52 at Yelahanka Sub-station has encountered a critical QoS/congestion event, as indicated by the Major Alarm ALARM_8979. This failure domain is rooted in the complex interplay between the node's role as an access device, its vendor-specific implementation of QoS protocols, and the inherent characteristics of the underlying hardware. Specifically, microbursts and interface tail drops can occur when the node's QoS queues become congested, leading to a starvation of critical traffic classes. This, in turn, can be exacerbated by the node's reliance on Huawei's proprietary QoS dialect, which may not always prioritize traffic classes in a predictable or consistent manner. Furthermore, the node's access role means that it is often the first point of contact for user traffic, making it particularly susceptible to congestion events. To address this issue, it is essential to identify the root cause of the QoS/congestion event and implement targeted resolution steps to mitigate its impact.

Pre-requisite Checks

- Verify that the node is powered on and all interfaces are online.
- Check the redundant link status to ensure that it is functioning correctly.
- Dispatch remote hands to Yelahanka Sub-station to perform physical checks, if necessary.
- Verify that the node's configuration is up-to-date and aligned with the latest Huawei recommendations.

Potential Root Causes

Cause	Probability
QoS queue misconfiguration	High
Interface congestion due to high traffic loads	Medium
Huawei QoS dialect implementation issues	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue statistics, Expected Output: Queue statistics for all QoS classes, Description: This command provides detailed statistics on QoS queue utilization, including dropped packets and queue depths. It helps identify potential bottlenecks and congestion points.

Command: display interface congestion statistics, Expected Output: Congestion statistics for all interfaces, Description: This command provides detailed statistics on interface congestion, including packet
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```
drops and interface utilization. It helps identify potential
congestion points and high-traffic interfaces.

Command: display qos class statistics, Expected Output: QoS class
statistics for all traffic classes, Description: This command provides
detailed statistics on QoS class utilization, including packet drops
and class utilization. It helps identify potential QoS class
misconfiguration and class starvation.
```

Resolution Steps

Step 1: Step: Configure QoS queues for optimal performance, Command: config qos queue priority , Description: This command adjusts the priority of a specific QoS queue to optimize its performance and prevent queue starvation.

Step 2: Step: Adjust QoS class configurations, Command: config qos class priority , Description: This command adjusts the priority of a specific QoS class to optimize its performance and prevent class starvation.

Step 3: Step: Increase interface bandwidth or optimize interface configurations, Command: config interface bandwidth , Description: This command increases the bandwidth of a specific interface or optimizes its configuration to prevent congestion and packet drops.

Post-Check Verification

```
Command: display qos queue statistics, Expected Output: Queue
statistics for all QoS classes (improved), Description: This command
verifies that QoS queue statistics have improved after implementing
resolution steps.

Command: display interface congestion statistics, Expected Output:
Congestion statistics for all interfaces (improved), Description: This
command verifies that interface congestion statistics have improved
after implementing resolution steps.

Command: display qos class statistics, Expected Output: QoS class
statistics for all traffic classes (improved), Description: This
command verifies that QoS class statistics have improved after
implementing resolution steps.
```

Roll-back Procedures

- Step: Restore original QoS queue configurations, Command: undo config qos queue , Description: This command restores the original QoS queue configurations to their previous state.

- Step: Restore original QoS class configurations, Command: undo config qos class , Description: This command restores the original QoS class configurations to their previous state.

- Step: Restore original interface configurations, Command: undo config interface , Description: This command restores the original interface configurations to their previous state.